Brooke D. Simmons

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Professional Summary

- Principal interests: galaxy evolution, supermassive black holes, black hole-galaxy co-evolution, change-detection in astrophysics and earth observation, pattern detection, data science
- Expert in citizen/community science as a facility for data analysis and discovery
- Expert on parametric image decomposition, quantified visual morphology, multi-wavelength AGN analysis
- Active Collaborations: COSMOS, CANDELS, 4MOST-TiDES & AGN, LS4, Zooniverse (Transient Group and Humanitarian leads), Galaxy Zoo (Deputy PI), Galaxy Zoo Bar Lengths (PI), The Planetary Response Network (PI)

Employment

Aug 2024 – present	Professor, Lancaster University
Aug 2021 – Aug 2024	Reader (equivalent to Associate Professor), Lancaster University
Sep 2018 – Aug 2021	Lecturer (equivalent to Assistant Professor), Lancaster University
Dec 2015 – Sep 2018	Einstein Fellow, UC San Diego
Jun 2012 – Dec 2015	Postdoctoral Researcher, University of Oxford

Education

2003 – 2007;	Ph.D., Astronomy, Yale University — Advisor: C. Megan Urry
2011 – 2012	*Black Hole Growth and Host Galaxy Co-Evolution Over 8 Billion Years of Cosmic Time*
2007 – 2011	On leave (family reasons)
2002 – 2003	M.S. & M. Phil, Astronomy, Yale University
2001	A.B., Astrophysical Sciences, Princeton University (honors)

Awards

Apr 2026	Advancing Research Impact in Society (ARIS) 2026 award: Zooniverse (team award)
Feb 2021	UKRI Future Leaders Fellowship
Jan 2019	Royal Astronomical Society Group Achievement Award: Galaxy Zoo (team award)
Dec 2015	Einstein Fellowship
Jan 2014	Henry Skynner Junior Research Fellowship, Balliol College, Oxford
Oct 2012	Junior Research Fellowship, Worcester College, Oxford
Jun 2012	James Martin Fellowship, Oxford Martin School, Oxford

Invited Talks

Prof. Simmons is regularly invited to give astrophysics seminars & colloquia. Other recent invited talks include:

Apr 2026	"Multidisciplinary Insights from Machine Learning and Citizen Science", Leverhulme Interdisciplinary Network on Algorithmic Solutions meeting, Belfast
Sep 2024	"Insights Into Galaxies, Supermassive Black Holes and Beyond with Citizen Science and Machine Learning", Royal Society-Academy of Science of South Africa meeting, Johannesburg
Sep 2021	"Machine Learning in Disaster Management", Alan Turing Institute, London
Feb 2019	"Citizen Science and the UN Sustainable Development Goals", Amnesty International, London
Oct 2018	"Improving Algorithms with the Zooniverse", Machine Learning and Data Sharing to Combat the Illegal Wildlife Trade, Zoological Society of London

Funding Awarded

as PI

Feb 2021	UKRI: "Leading the Next Generation of Data-Driven Discoveries", £1.5M
Jan 2020	BBSRC: "Innovative Digital Citizen Science: Active Learning for Disaster Relief", £20,000
Jun 2019	STFC-IAA/Lancaster: "Astrophysical Analysis Tools for Disaster Relief & Resilience", £22,000
Apr 2019	STFC: "Crowdsourcing and Machine Learning for Disaster Relief and Resilience", £212,000

Prior to joining Lancaster: 4 NASA awards (US$649,000 combined), 1 ESA award (€175,000)

Telescope Proposals Awarded *as PI*

Telescope proposals awarded include 101 orbits on the Hubble Space Telescope (GO-14606), 1 night on Keck/KCWI, 7.5 hr of Band A on Gemini-S/GMOS, 15 hr on IRAM-30m, and 34 nights combined on multiple 2-4m class telescopes.

Teaching

Prof. Simmons is an HEA Fellow (awarded by AdvanceHE in the UK). Specific teaching duties include:

Mar 2019 – present	Course Convenor, multiple course modules (Lancaster University)
Oct 2012 – Dec 2015	Tutor, 4th-year Astrophysics (University of Oxford)
Jan 2008 – Dec 2010	Professional tutor in math, science, reading, writing
Sep 2001 – Jan 2005	Teaching Fellow, multiple course modules (Yale University)
Jan 2001 – May 2001	Teaching Assistant, "The Universe" (Princeton University)

Student Supervision

Jason Shingirai Makechemu	PhD submission expected April 2028
Matthew Thorne	PhD, 2026
Izzy Garland	PhD, 2024; now Astrophysics Fellow at Masaryk University
David O'Ryan	PhD, 2024; now ESA Fellow (ESAC)

Dr. Simmons has also supervised 5 MSc students (J. Craig, A. Mohammed, M. Fahey, J Andrew, H Masters), >20 MPhys students, and >10 summer undergraduate students. The latter categories include A. Anil, J. Butterworth, H. Child, R. Cochrane, E. Day, S. Dicker, A. Griffin, A. Han, T. Hutchinson-Smith, B. Kushkuley, C. Lawson, T. Melvin, T Mummert, A. Schooneveld, J. Shanahan, M. Silcock, J. Smith, A. Tapia, E. Walls.

Postdoctoral and Specialist Supervision

Lydia Makrygianni	(PDRA) December 2023 – present
Alice Mead	(Disaster Management Specialist) December 2023 – present
Klaas Wiersema	(PDRA) October 2021 – May 2023 (now Senior Lecturer, U. Hertfordshire)
Danil Kuzin	(PDRA) September 2019 – Mar 2024 (now PDRA in Ecoacoustics AI, Oxford)

Public Dialogue

Prof. Simmons regularly gives invited public talks and contributes to blogs and social media.

Select Media Coverage

The Guardian, "Volunteers Worldwide Aided Rescue Efforts After Dorian"	*25th Oct, 2019*
Nature, "Citizen scientists aid Ecuador earthquake relief"	*3rd May, 2016*
Nature, "Crisis Mappers Turn To Citizen Scientists"	*19th Nov, 2014*
Sky & Telescope, "Citizen Scientists Probe Early Galaxies"	*29th Sep, 2014*

Interdisciplinary Roles

Founder, Zooniverse Analysis Group

A data science collaboration; partnerships in Computer Science, Economics, Information Science, Ecology

Founder, Zooniverse Transient Group

Detection of changes in time-series data, with particular emphasis on machine-human classification systems

Principal Investigator, The Planetary Response Network

Crisis response. Current and previous partners: Rescue Global, Oxford Machine Learning Research Group, ESA, Planet, Logistics Cluster

Prof. Simmons' interdisciplinary work in machine learning and disaster relief was highlighted in a UK report by Fuller (2018): Fuller, I., 2018, "World-Class Innovation: Outputs from STFC Frontier Research, Volume 2", Swindon, Science and Technology Facilities Council (pp. 12-13).

Refereed Publications

Note: candidate, supervised, and citizen scientist collaborator names in bold

Citizen science collaborator co-authors: E. Baeten, S. Beer, C. Macmillan, K. J. Jek, M. L. Peck, J. Tate, J. Wilcox

Publications as Lead or Supervising Author

As a permanent academic, it is my policy that papers from my group will be led by a student or postdoc wherever possible.

16. "Galaxy Zoo Bar Lengths: A Catalogue of Measurements from *Hubble Space Telescope* Images and the Evolution of Galactic Bar Structure at $z < 1$",
    T. Hutchinson-Smith, **B. D. Simmons**, K. L. Masters, A. Coil, I. Garland, T. Géron, S. Kruk, C. Lintott, R. Smethurst, **A. Tapia**, K. Willett, **E. Baeten, S. Beer, M. L. Peck, J. Wilcox**, submitted
15. "Quantifying the Maximum Recoverable Variance in Star Formation Rate from 1.4 GHz Radio Continuum Emission", **J. S. Makechemu,** J. O. Chibueze, **B. D. Simmons**, submitted
14. "Timescales for the Effects of Interactions on Galaxy Properties and SMBH Growth"
    D. O'Ryan, B. D. Simmons, A. Faisst, **I. L. Garland,** T. Géron, G. Gozaliasl, S. Gillman, **S. Pinto**, W. C. Keel, A. Koekemoer, S. Kruk, K. L. Masters, **O. Montoya C., M. Redden, M. Thorne, E. Walls, D. Weerasinghe,** J. Weaver, 2025, MNRAS, 539, 2967-2986
13. "Structural Decomposition of Merger-Free Galaxies Hosting Luminous AGNs"
    M. J. Fahey, I. L. Garland, **B. D. Simmons**, W. C. Keel, **J. Shanahan,** A. Coil, E. Glikman, C. J. Lintott, K. L. Masters, E. Moran, R. J. Smethurst, T. Géron, **M. Thorne**, 2025, MNRAS, 537, 3511-3524
12. "Galaxy Zoo DESI: large-scale bars as a secular mechanism for triggering AGN"
    I. L. Garland, M. Walmsley, **M. Silcock**, **L. Potts**, **J. Smith**, **B. D. Simmons**, C. Lintott, R. J. Smethurst, J. Dawson, W. C. Keel, K. B. Mantha, K. L. Masters, **D. O'Ryan**, J. Popp, **M. Thorne**, 2024, MNRAS, 532, 2320-2330
11. "Harnessing the Hubble Space Telescope Archives: A Catalogue of 21,926 Interacting Galaxies"
    D. O'Ryan, B. Merín, **B. D. Simmons,** A. Vojteková, A. Anku, M. Walmsley, **I. L. Garland**, T. Géron, W. C. Keel, S. Kruk, C. J. Lintott, K. B. Mantha, K. L. Masters, J. Reerink, R. J. Smethurst, and **M. Thorne**, 2023, ApJ, 948, 1, 40 (22 pp.)
10. "The most luminous, merger-free AGN show only marginal correlation with bar presence"
    I. Garland, M. Fahey, B. D. Simmons, R. J. Smethurst, C. J. Lintott, **J. Shanahan, M. Silcock, J. Smith,** W. C. Keel, A. Coil, T. Géron, K. L. Masters, **D. O'Ryan, M. R. Thorne,** and **K. Wiersema**, 2023, MNRAS, 522, 211-225
9. "Disaster, Infrastructure and Participatory Knowledge: The Planetary Response Network"
    B. D. Simmons, *et al.* (15 authors), 2022, Citizen Science: Theory & Practice, 7(1): 21, pp. 1-16
8. "Disaster mapping from satellite images: damage detection from crowdsourced point labels"
    D. Kuzin, O. Isupova, **B. D. Simmons,** C. Lintott, S. Reece, 2021, in 3rd Workshop on Artificial Intelligence for Humanitarian Assistance and Disaster Response (NeurIPS 2021). arxiv:2111.03693 (10 pp.)
7. "Supermassive black holes in disk-dominated galaxies outgrow their bulges and co-evolve with their host galaxies"
    B. D. Simmons, R. J. Smethurst, and C. Lintott, 2017, MNRAS, 470, 1559-1569
6. "Galaxy Zoo: Quantitative Visual Morphological Classifications for 48,000 galaxies from CANDELS"
    B. D. Simmons, C. Lintott, K. W. Willett, K. L. Masters, *et al.* (46 authors), 2017, MNRAS, 464, 4420-4447
5. "Galaxy Zoo: CANDELS Barred Disks and Bar Fractions"
    B. D. Simmons, **T. Melvin**, C. Lintott, K. L. Masters, *et al.* (**K. J. Jek**: 36th of 42 authors), 2014, MNRAS, 445, 3466-3474
4. "Galaxy Zoo: Bulgeless Galaxies With Growing Black Holes"
    B. D. Simmons, *et al.* (**A. Han**: 5th of 11 authors), 2013, MNRAS, 429, 2199-2211
3. "Moderate-luminosity Growing Black Holes from $1.25 < z < 2.7$: Varied Accretion in Disk-Dominated Hosts"
    B. D. Simmons, C. M. Urry, K. Schawinski, C. Cardamone, and E. Glikman, 2012, ApJ, 761, 75 (10 pp.)
2. "Obscured GOODS AGN and Their Host Galaxies at z < 1.25: The Slow Black Hole Growth Phase"
    B. D. Simmons, J. Van Duyne, C. M. Urry, E. Treister, A. M. Koekemoer, N. A. Grogin, and the GOODS Team, 2011, ApJ, 734, 121 (16 pp.)
1. "The Accuracy of Morphological Decomposition of Active Galactic Nucleus Host Galaxies"
    B. D. Simmons and C. M. Urry, 2008, ApJ, 683, 644-658

Publications in Zooniverse Analysis Group and Zooniverse Transients Group roles

15. "Brightness, Colour, Polarisation: A Multi-Instrument Observation Campaign of the Gorizont-6 Satellite" Airey, R. J. S. *et al.* (**Simmons**: 17th of 18 authors), 2026, ASR., 77, 12, 12055-12068
14. "Short timescale imaging polarimetry of geostationary satellite Thor-6: the nature of micro-glints" **K. Wiersema** *et al.* (**Simmons**: 10th of 11 authors), 2022, ASR, 70, 10, 3003-3015
13. "Camera settings and biome influence the accuracy of citizen science approaches to camera trap image classification", N. Egna *et al.* (**Simmons**: 7th of 26 authors), 2020, Ecology & Evolution, 10, 21, p. 11954-11965
12. "Rapid Post Disaster Damage Mapping with Satellite Imagery, Citizen Scientists and Machine Learning" O. Isupova, D. Kuzin, **B. D. Simmons**, S. Reece, 2019, DwD-GCRF-UKADR-DRG-UKCDR International Conference, 2019
11. "Everyone counts? Design considerations in online citizen science" H. Spiers, *et al.* (**Simmons**: 4th of 7 authors), 2019, JCOM, 18(01), A04 (32 pp.)
10. "Getting Connected: An Empirical Investigation of the Relationship Between Social Capital and Philanthropy Among Online Volunteers", J. Cox, E. Y. Oh, **B. D. Simmons**, *et al.* (8 authors), 2018, NVSQ, 48(2), 1-23
9. "Exposing the Science in Citizen Science: Fitness to Purpose and Intentional Design", J. K. Parrish, H. Burgess, J. F. Weltzin, L. Fortson, A. Wiggins, **B. D. Simmons**, 2018, Integr. Comp. Biol., icy032 (11 pp.)
8. "Integrating Human And Machine Intelligence In Galaxy Morphology Classification Tasks", M. Beck, *et al.* (**Simmons**: 7th of 11 authors), 2018, MNRAS, 476, 5516-5534
7. "The K2-138 System: A Near-Resonant Chain of Five Sub-Neptune Planets Discovered by Citizen Scientists", J. L. Christiansen, *et al.* (**Simmons**: 6th of 27 authors), 2018, AJ, 155, 57 (9 pp.)
6. "Doing good online: The changing relationships between motivations, activity and retention among online volunteers", J. Cox, E. Y. Oh, **B. D. Simmons**, *et al.* (8 authors), 2018, NVSQ, 47(5), 1031-1056
5. "A transient search using combined human and machine classifications" D. Wright, *et al.* (**Simmons**: 18th of 26 authors), 2017, MNRAS, 472, 1315-1323
4. "Assessing Data Quality In Citizen Science" M. Kosmala, A. Wiggins, A. Swanson, **B. D. Simmons**, 2016, Front. Ecol. Environ., 14(10): 551–560
3. "Science Learning via Participation in Online Citizen Science" K. L. Masters, E. Y. Oh, J. Cox, *et al.* (**Simmons**: 4th of 8 authors), 2016, JCOM, 15(03), A07 (33 pp.)
2. "Playing with science: Exploring how game activity motivates users' participation on an online citizen science platform", A. Greenhill, *et al.* (**Simmons**: 5th of 9 authors), 2015, Aslib Jour. Info. Mgmt, 68, 306-325
1. "Defining and measuring success in online citizen science: A case study of Zooniverse projects" J. Cox, E-Y. Oh, **B. D. Simmons**, *et al.* (8 authors), 2015, CISE, 17, 28-41

Publications as Major Contributing Author

41. "Evidence for secularly-driven galaxy-SMBH co-evolution in 2,435 disk DESI galaxies", S. M. Jewell, R. J. Smethurst, C. J. Lintott, **B. D. Simmons,** R. S. Beckmann, I. L. Garland, T. Géron, B. W. Holwerda, R. Pucha, and the Galaxy Zoo team, submitted (20 pp.)
40. "The Complex Relationships Between AGN, Bars and Bulges", I. Garland, *et al.* (**B. D. Simmons:** 7th of 12 authors), 2026, A&A, 709, A48 (9 pp.)
39. "Galaxy Zoo: Cosmic Dawn - morphological classifications for over 41 000 galaxies in the Euclid Deep Field North from the Hawaii Two-0 Cosmic Dawn survey", J. Pearson, *et al.* (**B. D. Simmons:** 8th of 25 authors), 2025, MNRAS, 546, 3, id.staf2250 (23 pp.)
38. "Euclid Quick Data Release (Q1): First visual morphology catalogue", Euclid Collaboration; M. Walmsley, *et al.* (**B. D. Simmons, J. S. Makechemu, E. M. Baeten, C. Macmillan:** 14th, 15th, 22nd, and 23rd of 354 authors), submitted
37. "Euclid Quick Data Release (Q1), A first look at the fraction of bars in massive galaxies at $z < 1$", Euclid Collaboration; M Huertas-Company, *et al.* (**J. S. Makechemu, B. D. Simmons**: 20th and 21st of 338 authors), submitted
36. "Galaxy Zoo CEERS: Bar fractions up to $z \sim 4.0$", T. Géron, *et al.* (**J. S. Makechemu, B. D. Simmons**: 8th and 13th of 25 authors), 2025, ApJ, 987, 74 (19 pp.)
35. "Galaxy Zoo JWST: Up to 75% of discs are featureless at $3 < z < 7$", R. J. Smethurst, **B. D. Simmons**, *et al.* (**J. S. Makechemu, M. R. Thorne**: 10th and 15th of 24 authors), 2025, MNRAS, 539, 2, pp. 1359-1371

34. “Supermassive black holes in merger-free galaxies have higher spins which are preferentially aligned with their host galaxy”, R. Beckmann, R. J. Smethurst, **B. D. Simmons,** A. Coil, Y. Dubois, **I. L. Garland**, C. J. Lintott, G. Martin, S. Peirani, and C. Pichon, 2024, MNRAS, 527, 4, pp. 10867-10877
33. “Evidence for non-merger co-evolution of galaxies and their supermassive black holes”, R. J. Smethurst, R. Beckmann, **B. D. Simmons,** A. Coil, J. Devriendt, Y. Dubois, **I. L. Garland**, C. J. Lintott, G. Martin, and S. Peirani, 2024, MNRAS, 527, 4, pp. 10855-10866
32. “Galaxy Zoo DESI: Detailed morphology measurements for 8.7M galaxies in the DESI Legacy Imaging Surveys”, M. Walmsley, et al. (**I. L. Garland, D. O’Ryan, B. D. Simmons, E. Baeten, C. Macmillan**: 10th, 12th, 14th , 15th and 16th of 16 authors), 2023, MNRAS, 523, 4768-4786
31. “Searching for the Role of Mergers in Fast and Early SMBH Growth: Morphological Decomposition of Quasars and Their Hosts at z ~ 4.8”, M. O. Thomas, O. Shemmer, B. Trakhtenbrot, P. Lira, H. Netzer, **B. D. Simmons**, and N. Ilan, 2023, ApJ, 955, 1, 15 (17 pp.)
30. “A Dual QSO At Cosmic Noon”, E. Glikman, R. Langgin, M. A. Johnstone, I. Yoon, J. M. Comerford, **B. D. Simmons**, M. Lacy, and J. M. O’Meara, 2023, ApJL, 951, 1, L18 (8 pp.)
29. “Gems of the Galaxy Zoos - a Wide-Ranging *Hubble Space Telescope* Gap-Filler Program”, W. C. Keel, et al. (**B. D. Simmons**, **J. Shanahan, I. L. Garland** and **D. O’Ryan**: 7th, 12th, 14th and 16th of 16 authors), 2022, AJ, 163, 150 (17pp)
28. “Quantifying the Poor Purity and Completeness of Morphological Samples Selected by Galaxy Colour”, R. J. Smethurst, K. L. Masters, **B. D. Simmons**, et al. (**I. L. Garland** and **D. O’Ryan**: 4th and 9th of 10 authors), 2022, MNRAS, 510, 4126-4133
27. “Galaxy Zoo DECaLS: Detailed Visual Morphology Measurements from Volunteers and Deep Learning for 314,000 Galaxies”, M. Walmsley, et al. (**Simmons**, **E. Baeten** and **C. Macmillan**: 14th, 17th and 18th of 18 authors), 2022, MNRAS, 509, 3966-3988
26. “Kiloparsec-scale AGN Outflows and Feedback in Merger-Free Galaxies”, R. J. Smethurst, **B. D. Simmons**, et al. (**J. Shanahan** and **I. L. Garland**: 9th and 10th of 10 authors), 2021, MNRAS, 507, 3985-3997
25. “Galaxy Zoo Builder: Four-Component Photometric decomposition of Spiral Galaxies Guided by Citizen Science”, T. Lingard, et al. (**Simmons** and **E. Baeten**: 6th and 10th of 10 authors), 2020, ApJ, 900, 2 (18 pp.)
24. “Galactic Conformity in both Star Formation and Morphological Properties”, J. A. Otter, K. L. Masters, **B. D. Simmons**, C. J. Lintott, 2020, MNRAS, 492, 2722-2730
23. “Secularly powered outflows from AGN: the dominance of non-merger driven supermassive black hole growth”, R. J. Smethurst, **B. D. Simmons**, C. J. Lintott, **J. Shanahan**, A. Coil, W. C. Keel, E. Glikman, E. C. Moran, K. L. Masters, C. M. Urry, K. W. Willett, 2019, MNRAS, 489, 4016-4031
22. “SNITCH: Seeking a simple, informative star formation history inference tool”, R. J. Smethurst, M. Merrifield, C. J. Lintott, K. L. Masters, **B. D. Simmons**, *et al.* (10 authors), 2019, MNRAS, 484, 3590-3603
21. “Galaxy Zoo: constraining the origin of spiral arms”, R. Hart, S. Bamford, W. C. Keel, S. Kruk, K. L. Masters, **B. D. Simmons**, R. J. Smethurst, 2018, MNRAS, 478, 932-949
20. “Normal black holes in bulge-less galaxies: the largely quiescent, merger-free growth of black holes over cosmic time”
    G. Martin, S. Kaviraj, M. Volonteri, **B. D. Simmons**, *et al*. (9 authors), 2018, MNRAS, 476, 2801-2812
19. “Galaxy Zoo: Secular evolution of barred galaxies from structural decomposition of multi-band images”
    S. Kruk, C. Lintott, S. Bamford, K. Masters, **B. D. Simmons**, *et al.* (12 authors), 2018, MNRAS, 473, 4731-4753
18. “Galaxy Zoo: Finding offset discs and bars in SDSS galaxies”
    S. J. Kruk, C. J. Lintott, **B. D. Simmons**, S. Bamford, *et al.* (12 authors), 2017, MNRAS, 469, 3363-3373
17. “Morphology and the Color-Mass Diagram As Clues to Galaxy Evolution at z ~ 1”
    M. C. Powell, C. M. Urry, C. Cardamone, **B. D. Simmons**, *et al.* (7 authors), 2017, ApJ, 835, 22 (10 pp.)
16. “Galaxy Zoo: Morphological Classifications for 120,000 Galaxies in HST Legacy Imaging”
    K. W. Willett, *et al.* (**Simmons**: 7th & **Han**: 15th & **Melvin:** 17th of 21 authors), 2016, MNRAS, 464, 4176-4203
15. “Galaxy Zoo: Evidence for rapid, recent quenching across a population of AGN host galaxies”
    R. J. Smethurst, C. Lintott, **B. D. Simmons**, *et al.* (11 authors), 2016, MNRAS, 463, 2986-2996
14. “Major Mergers Host the Most Luminous Red Quasars at *z* ~ 2: A *Hubble Space Telescope* WFC3/IR Study”
    E. Glikman, **B. D. Simmons**, M. Mailly, K. Schawinski, C. M. Urry, M. Lacy, 2015, ApJ, 806, 218 (24 pp.)
13. “Radio Galaxy Zoo: host galaxies and radio morphologies derived from visual inspection”

J. Banfield, *et al.* (**Simmons**: 7th of 36 authors), 2015, MNRAS, 453, 2326-2340

12. "Galaxy Zoo: Evidence For Diverse Star Formation Histories Through The Green Valley"
    R. J. Smethurst, C. J. Lintott, **B. D. Simmons**, *et al.* (13 authors), 2015, MNRAS, 450, 435-453
11. "Galaxy Zoo: The dependence of the star formation-stellar mass relation on spiral disk morphology"
    K. W. Willett, K. Schawinski, **B. D. Simmons**, *et al.* (**Melvin**: 7th of 13 authors), 2015, MNRAS, 449, 820-827
10. "The Green Valley is a Red Herring: Galaxy Zoo reveals two evolutionary pathways towards quenching of star formation in early- and late-type galaxies"
    K. Schawinski, C. M. Urry, **B. D. Simmons**, *et al.* (15 authors), 2014, MNRAS, 440, 889-907
9. "Galaxy Zoo: Evolution of the bar fraction over the last eight billion years from HST-COSMOS"
    T. Melvin, *et al.* (**Simmons:** 5th of 14 authors), 2014, MNRAS, 438, 2882-2897
8. "Galaxy Zoo 2: detailed morphological classifications for 304,122 galaxies from the Sloan Digital Sky Survey"
    K. W. Willett, *et al.* (**Simmons:** 5th & **Melvin**: 11th of 18 authors), 2013, MNRAS, 435, 2835-2860
7. "Major Galaxy Mergers Only Trigger the Most Luminous AGN"
    E. Treister, K. Schawinski, C. M. Urry, and **B. D. Simmons**, 2012, ApJL, 758, 39 (5 pp.)
6. "Heavily Obscured Quasar Host Galaxies at $z \sim 2$ are Disks, Not Major Mergers"
    K. Schawinski, **B. D. Simmons**, C. M. Urry, E. Treister, and E. Glikman, 2012, MNRAS Letters, 425, L61-L65
5. "Bolometric Luminosities and Eddington Ratios of X–ray Selected AGN in the XMM-COSMOS Survey"
    E. Lusso, A. Comastri, **B. D. Simmons**, *et al.* (27 authors), 2012, MNRAS, 425, 623-640
4. "Evidence for Three Accreting Black Holes in a Galaxy at z ~ 1.35: A Snapshot of Recently Formed Black Hole Seeds?" K. Schawinski, C. M. Urry, E. Treister, **B. D. Simmons**, P. Natarajan, and E. Glikman, 2011, ApJL, 743, 37 (6 pp.)
3. "HST WFC3/IR Observations of AGN Hosts at z ~ 2: Supermassive Black Holes Grow in Disk Galaxies"
    K. Schawinski, E. Treister, C. M. Urry, C. N. Cardamone, **B. D. Simmons**, and S. K. Yi, 2011, ApJL, 727, 31 (6 pp.)
2. "Do Moderate-Luminosity Active Galactic Nuclei Suppress Star Formation?"
    K. Schawinski, **B. D. Simmons**, *et al.* (**B. Kushkuley**: 7th of 7 authors), 2009, ApJL, 692, L19-L23
1. "Active Galactic Nucleus Host Galaxy Morphologies in COSMOS"
    J. M. Gabor, C. D. Impey, K. Jahnke, **B. D. Simmons**, *et al.* (17 authors), 2009, ApJ, 691, 705-722

Team Publications as Contributing Author

48. "Deep learning segmentation of spiral arms and bars for 600,000 galaxies in DESI", A. Spindler, *et al.* (Simmons: 6th of 7 authors), submitted
47. "Here Be SDRAGNs — Spiral Galaxies Hosting Large Double Radio Sources", Tate, J., *et al.* (Simmons: 15th of 19 authors), 2026, ApJ, 171, 289 (34 pp.)
46. "The effects of bar strength and kinematics on galaxy evolution II: The global and local impacts of slow-strong bars", P. Mengistu, *et al.* (Simmons: 6th of 6 authors), 2026, MNRAS, 548, 2, id.stag561 (19 pp.)
45. "The La Silla Schmidt Southern Survey", A. Miller, *et al.* (Simmons: 64th of 78 authors), 2025, PASP, 137, id 094204 (30 pp.)
44. "TiDES: The 4MOST Time Domain Extragalactic Survey", C. Frohmaier, *et al.* (Simmons: 23rd of 27 authors), 2025, ApJ, 992, 158 (18 pp.)
43. "COSMOS-Web: The emergence of the Hubble Sequence", M. Huertas-Company, *et al.* (Simmons: 29th of 32 authors), 2025, A&A, 704, A94 (20 pp.)
42. "Radio Galaxy Zoo data release 1: 100185 radio source classifications from the FIRST and ATLAS surveys", O. I. Wong, *et al.* (Simmons: 17th of 18 authors), 2025, MNRAS, 536, 3488-3506
41. "The Prevalence of Star-forming Clumps as a Function of Environmental Overdensity in Local Galaxies", D. Adams, *et al.* (Simmons: 9th of 10 authors), 2025, ApJ, 979, 118 (15 pp.)
40. "Through the Citizen Scientists' Eyes: Insights into Using Citizen Science with Machine Learning for Effective Identification of Unknown-Unknowns in Big Data", K. B. Mantha, *et al.* (Simmons: 9th of 14 authors), 2024, CSTP, 9(1): 40, pp. 1–15
39. "The effects of bar strength and kinematics on galaxy evolution: slow strong bars affect their hosts the most", T. Géron, *et al.* (Simmons: 8th of 8 authors), 2024, ApJ, 973, 129 (24 pp.)

38. "Transfer learning for galaxy feature detection: Finding giant star-forming clumps in low redshift galaxies using Faster Region-based Convolutional Neural Networks", J. Popp, *et al.* (Simmons: 11th of 11 authors), 2024, RASTI, 3, 1, pp. 174-197

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